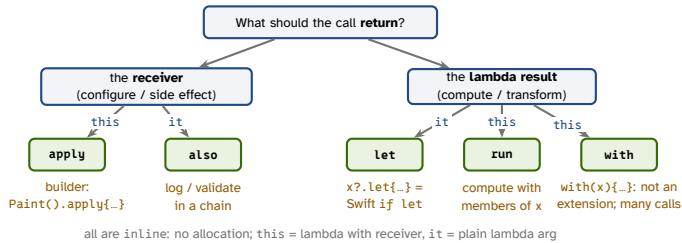
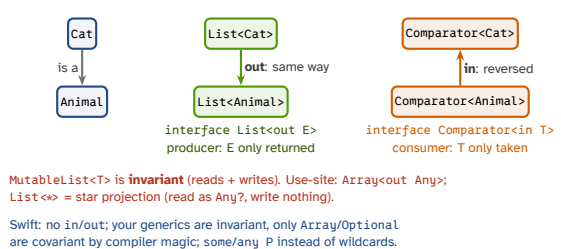


In one line: Kotlin is Swift's closest cousin: **nullability in the type system**, **immutable-by-default bindings**, **sealed hierarchies** + **exhaustive when**, extensions and trailing lambdas. The differences that matter: **classes only** (no value-type structs — data class is a reference type), **JVM generics are erased** (reified is the escape), extensions are **statically dispatched**, and internal means **Gradle module**.

Scope functions — which one?



Declaration-site variance



Null safety & types

- String never null, String? may be. ?. safe call, ?: Elvis (right side may return/throw), !! = NPE if null (Swift !). Smart cast after if (x != null) / is — only for vals/locals the compiler can prove unchanged.
- Platform types** String!: Java values without nullness annotations — the compiler trusts you; a null blows up at *first use*, far from the cause.
- val = read-only reference (Swift let), *not* deep immutability. lateinit var: non-null, set later, throws if read first.
- == calls equals(), === = same instance. Unit = Void, Nothing = Never. No checked exceptions.

Classes

- final by default** — open to subclass. Visibility: public (default!), internal = same **module**, protected = subclasses, private top-level = file.
- data class**: equals/hashCode/toString/copy()/componentN() from the **primary constructor only** — body properties ignored. Still a *reference*: copy is shallow.
- sealed class/interface**: subclasses known at compile time (same module + package) → when without else; exhaustiveness is an error for sealed/enum/Boolean subjects. Guards: is A if x > 0 → ... (2.2). Swift enum with payloads ≈ sealed + data classes.
- @JvmInline value class**: one property, unboxed at runtime (boxed when nullable/generic/as interface) — a zero-cost type-safe id.
- object = thread-safe lazy singleton; companion object = statics.

Example

```
@JvmInline value class UserId(val raw: String)
data class User(val id: UserId, val name: String?)
sealed interface Load<out T> { // out: Load<Nothing> fits all
    data object Loading : Load<Nothing>
    data class Done<T>(val value: T) : Load<T>
    data class Failed(val e: Throwable) : Load<Nothing>
}
fun label(s: Load<User>): String = when (s) { // no else needed
    Load.Loading -> "... "
    is Load.Done -> s.value.name?.uppercase() ? "anon" // smart cast
    is Load.Failed -> s.e.message!! // NPE if null
}
inline fun <reified T> Any.asOrNull(): T? = this as? T // T kept
val cfg by lazy { loadConfig() } // once, SYNCHRONIZED
fun html(init: Html.() -> Unit) = Html().apply(init) // DSL
```

Functions, lambdas, delegation

- Extensions** compile to static functions taking the receiver: **resolved by the declared type**, never overridden; a **member always wins**. Nullable receivers OK (String?.isEmpty()).
- inline fun** pastes body + lambda into the call site: no lambda object; return in the lambda returns from the *caller* (non-local). **noinline** keeps a lambda as an object, **crossinline** bans non-local return. **reified T** (inline only) survives erasure: is T, T::class.
- Lambda with receiver T.() -> R**: this is a T — buildString, Gradle Kotlin DSL, Compose scopes; @DslMarker stops outer receivers leaking in.
- Delegation**: by lazy (SYNCHRONIZED default), Delegates.observable (≈ didSet), by map; **class delegation** class Logged(d: Repo) : Repo by d = free forwarding.

Swift ↔ Kotlin

Swift	Kotlin
struct (value, copy)	data class (reference; copy() by hand)
enum + associated values	sealed interface + data class/object
protocol + extension default	interface with default bodies
guard let x else { return }	val x = y ?: return
lazy var · didSet	by lazy · Delegates.observable
static · fileprivate	companion object · private top-level
internal (default)	internal (opt-in; default public)
@resultBuilder	lambda with receiver (DSL)
generics kept at runtime	erased on JVM; reified if inline

Versions: K2 compiler (new FIR frontend) stable + default since **Kotlin 2.0.0** (May 2024). Current **2.4.20** (Sep 2026); 2.4 made **context parameters** (context(log: Logger) fun ...) stable, 2.2 when guards and non-local break/continue.

Traps

- ! “data class is a struct” — no: shared reference, copy() shallow; a var in it breaks HashSet lookups after mutation.
- ! Overriding an extension — impossible; the static type decides.
- ! !! to silence the compiler — use ?:, ?. or requireNotNull(x).

Remember: nulls in types · final + public by default · extensions static · erased unless reified.